

Group name

Enter your details :

Section

Instructions/information: Following are some exercises that relate to a reading you have been assigned. Please refer to your notes as you answer them. You are encouraged to confer with your group mates as you put down your thoughts.

4.1 The Elements of Causal Graphs

4.1.1 Module 4 | Variables

Reflect and discuss (A) Causal Graphs: Variables 1

 1. _____ represent: (choose one)

- A. Causal relationships between particular events.
- B. Causal relationships between test pairs
- C. Causal relationships between variables.
- D. None of the above.

 2. Which of the following does a _____ contain? (choose one)

- A. Causal assignments and test pairs.
- B. The probability distribution over the variable designated as the effect.
- C. Variables and directed edges.
- D. A response structure.

4.1.2 Module 4 | Directed Edges

Reflect and discuss (B) Directed Edges 1

 3. A directed edge _____ in a causal graph means that: (choose one)

- A. X promotes the occurrence of Y, relative to the other variables in the system.
- B. Y always occurs when X does, relative to the other variables in the system.
- C. X is the only cause of Y, relative to the other variables in the system.
- D. X is a direct cause of Y, relative to the other variables in the system.

 4. If there is a _____ from X to Y in a causal graph, then this means that:

- A. Whenever X occurs, Y must occur also.
- B. There is at least one test pair of causal assignments for X in which there is a difference in Y.
- C. Y cannot occur except when X is present.
- D. Y is a direct cause of X.

4.1.3 Module 4 | Deterministic Causation: The Malaria Example

Reflect and discuss (C) Deterministic Causation: The Malaria Example 1

 5. Referring to the response structure in the table titled: Response Structure For The Malaria Case, which of the following are test pairs of causal assignments for _____? (Note: Not all test pairs are on this list.) (choose all that apply)

- A. Assignments 1 and 3.
- B. Assignments 2 and 6.
- C. Assignments 4 and 7.
- D. Assignments 9 and 13.

 6. If we built a causal graph of the system that includes: {Bitten, Inoculated, Gene, Drinker, Malaria}, based on the response structure in the table titled: Response Structure For The Malaria Case, should there be a directed edge from _____ to 'Malaria'? (choose one)

- A. Yes
- B. No

 7. Which of the following pairs of causal assignments should we compare to determine whether _____ is a direct cause of Malaria? (choose all that apply)

- A. Assignments 1 and 2.
- B. Assignments 15 and 16.
- C. Assignments 5 and 7.
- D. Assignments 11 and 12.

 8. If we built a causal graph of the system that includes: {Bitten, Inoculated, Gene, Drinker, Malaria}, should there be a directed edge from _____ to 'Malaria'? (choose one)

- A. Yes
- B. No

4.1.4 Module 4 | Indeterministic Causation: The Cell Phone Example

Reflect and discuss (D) Indeterministic Causation: The Cell Phone Example 1

 9. Referring to the response structure for CONNECTED shown in the table titled: Deterministic Response Structure For Connected, which of the following are test pairs for _____? (choose all that apply)

- A. 1 and 2.
- B. 1 and 3.
- C. 1 and 4.
- D. 2 and 3.
- E. 2 and 4.
- F. 3 and 4.

 10. In which of the following test pairs for CALL PLACED does _____ vary? (choose one)

- A. 1 and 3
- B. 2 and 4
- C. Neither.

 Reflect and discuss (E) Indeterministic Causation: The Cell Phone Example 2

Assignment	Call Placed	Battery Charged	In Range of Tower	Effect: Call Connected
1	Send	Yes	Yes	Yes
2	Send	Yes	No	No
3	Send	No	Yes	No
4	Send	No	No	No
5	End	Yes	Yes	No
6	End	Yes	No	No
7	End	No	Yes	No
8	End	No	No	No

 11. If we consider the causal system: {Call Placed, Battery Charged, In Tower Range, Connected}, with the response structure given above, Call Placed is a direct cause of Call Connected. If we consider the smaller system including only: {Call Placed, Battery Charged, Connected}, the response structure for _____ would be: (choose all that apply)

- A. Deterministic.
- B. Truly-Indeterministic.
- C. Psuedo-Indeterministic.
- D. None of the above.

 12. Again, relative to the smaller system including only: {Call Placed, Battery Charged, Connected}, _____ is a direct cause of Connected if: (choose all that apply)

- A. There is at least one test pair for Call Placed such that Connected changes from Yes to No, or from No to Yes.
- B. There is at least one test pair for Call Placed such that the probability of Connected changes.
- C. Neither of the above.

4.2 Representing Different Varieties of Causation

4.2.1 Module 4 | Causal Chains and Direct vs Indirect Causation

Reflect and discuss (F) Causal Chains 1

 13. What is the difference between a causal system in which there is a causal chain from x to y to z but in which x is not a direct cause of z, and another causal system in which there is a causal chain from x to y to z but in which x is also a direct cause of z?

Reflect and discuss (G) Causal Chains 2

 14. Consider the causal system: {Exposure (to other kids with Chicken Pox), Infected (with Chicken Pox Virus), Have Chicken Pox Rash}. Using your common sense, is _____ a direct cause of Have Chicken Pox Rash relative to this system? (choose one)

- A. Yes
- B. No
- C. Not enough information to tell

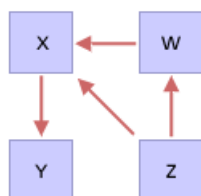
 15. Consider the smaller system: {Exposure (to other kids with Chicken Pox), Have Chicken Pox Rash}. Using your common sense, is _____ a direct cause of Have Chicken Pox Rash relative to this system? (choose one)

- A. Yes
- B. No

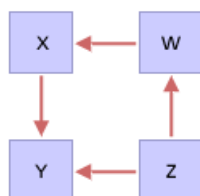
4.2.2 Module 4 | Cyclic Causal Graphs

Reflect and discuss (H) Cyclic Causal Graphs 1

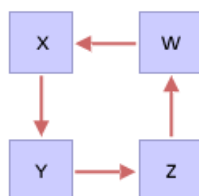
 16. Which of the following causal graphs contains ? (choose all that apply)



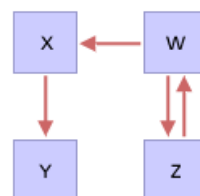
A.



B.



C.



D.

4.2.3 Module 4 | Case Studies

Reflect and discuss (I) Case Studies 1

 17. Consider the following list:

- A. Adult [yes, no].
- B. Exposed to Alcohol in the Womb [yes, no].
- C. Drug dependence [none, alcohol, cocaine, heroin, other].
- D. Has Ever Received Psychiatric Treatment [yes, no].
- E. Treatment for alcohol or drug dependence [yes, no].
- F. Anxiety or Eating Disorder [yes, no].

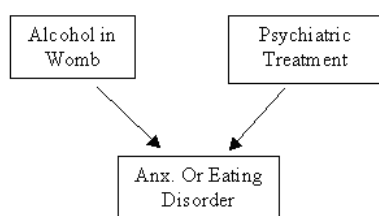
Which of the following best describes the main set of _____ discussed in the report on Fetal Exposure to Alcohol: (choose one)

- A. 1, 2 and 4.
- B. 2, 3 and 4.
- C. 2, 4 and 6.
- D. 3, 4 and 5.

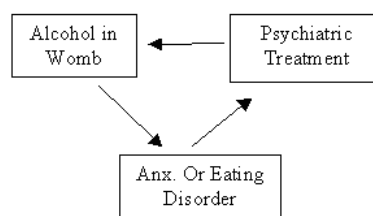
 18. Assume that the main variables in the study were: (choose one)

- A. Exposed to Alcohol in the Womb [yes, no]
- B. Has Ever Received Psychiatric Treatment [yes, no]
- C. Anxiety or Eating Disorder [yes, no]

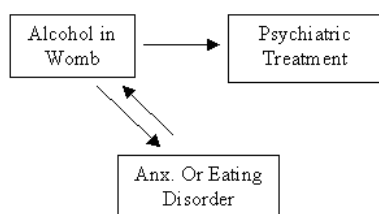
Which of the following is the causal graph that best captures the _____ discussed in the study:



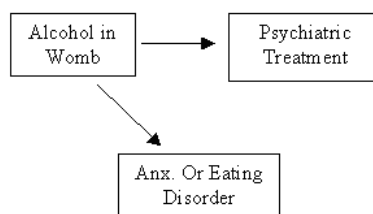
A.



B.



C.



D.

 Reflect and discuss (J) Case Studies 2

 19. Consider the following list:

- A. Elderly [yes, no]
- B. Exercise [3 times weekly, no]
- C. Body's Blood Pumping Capacity [poor, avg., good]
- D. Cognitive Function [poor, avg., good]
- E. Medicated for Depression [yes, no]
- F. Blood's Oxygen Carrying Capacity [poor, avg., good]

Which of the following best describes the _____ that were measured in the report on Exercise and Mental Decline: (choose one)

- A. 1, 2 and 4.
- B. 2, 3 and 5.
- C. 2, 4 and 6.
- D. 2, 4 and 5.

 20. Consider again the following list: (choose one)

- A. Elderly [yes, no]
- B. Exercise [3 times weekly, no]
- C. Body's Blood Pumping Capacity [poor, avg., good]
- D. Cognitive Function [poor, avg., good]
- E. Medicated for Depression [yes, no]
- F. Blood's Oxygen Carrying Capacity [poor, avg., good]

Which of the following best describes the _____ that were part of a mechanism for how exercise might improve cognitive function?

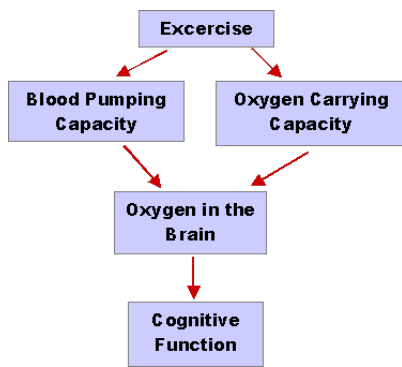
- A. 1, 2, 3 and 5.
- B. 2, 3, 4 and 5.
- C. 2, 3, 4 and 6.
- D. 2, 4, 5 and 6.

 21. Consider the following variables: (choose one)

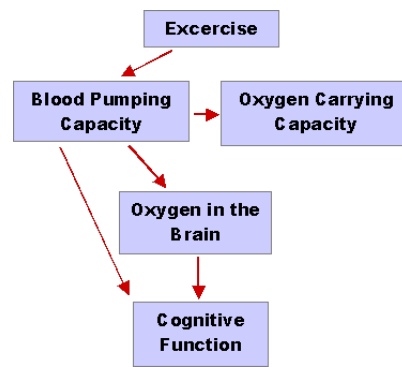
- Exercise [3 times weekly, no]
- Body's Blood Pumping Capacity [poor, avg., good]
- Blood's Oxygen Carrying Capacity [poor, avg., good]
- Oxygen Carrying Capacity [poor, avg., good]
- Cognitive Function [poor, avg., good]

Which of the following graphs best represents the _____ consisting of the above variables?

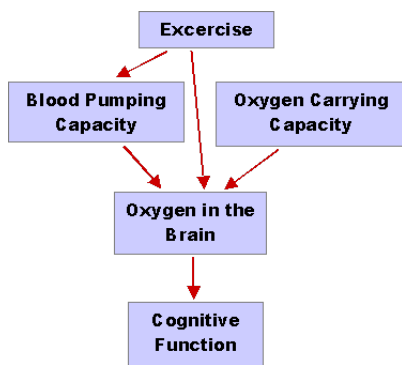
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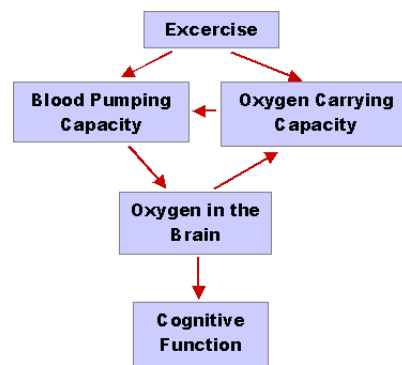
A.



B.



C.



D.